



AI-Powered US Tax & Legal Research System

Hybrid retrieval + Graph RAG over 100 US legal & tax documents —
grounded answers with exact citations.

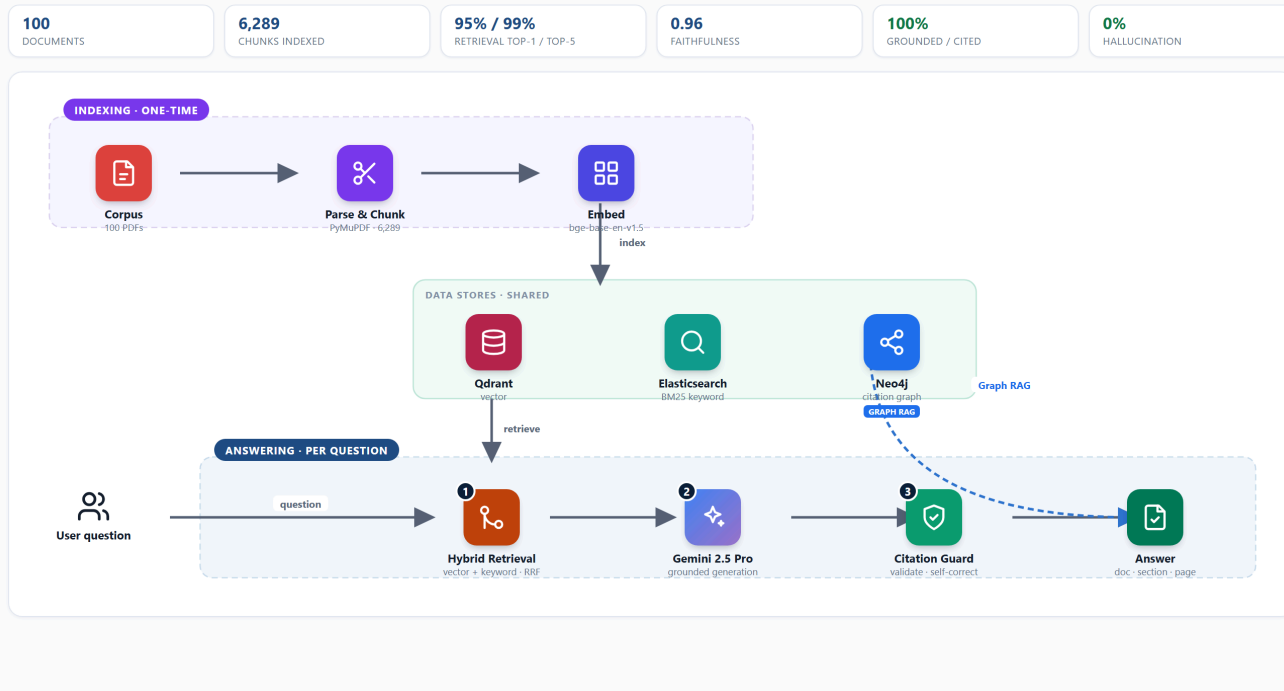
Live demo ↗ kuldeepmishra3-legal-rag.hf.space

Assignment submission · Kuldeep Mishra

Architecture

System Architecture — US Tax & Legal RAG

Documents are indexed once; each question is answered live with grounded, cited output.



Documents are indexed once (top lane) into a shared vector (Qdrant), keyword (Elasticsearch), and citation-graph (Neo4j) store. Each user question runs through hybrid retrieval (RRF), grounded generation with Gemini 2.5 Pro, and a citation-validation guard — producing an answer cited to document, section, and page. Neo4j additionally powers Graph RAG (relationship queries).

Evaluation Report

Evaluated end-to-end on a hand-authored **100-query golden set** (25 per category: Acts / Court Judgments / POV / Tax; mixed difficulty). Answer model: **Gemini 2.5 Pro**; correctness judge: Gemini 2.5 Flash.

1. Retrieval Accuracy

The system finds the correct source document at **95% Top-1** and **99% Top-5**.

Category	Top-1	Top-5
Acts	100%	100%
POV (commentary)	100%	100%
Court Judgments	96%	96%
Tax Documents	84%	100%
Overall	95%	99%

2. Faithfulness Score

- **Faithfulness (RAGAS): 0.96** — answers are almost entirely supported by the retrieved context.
- Context recall **0.98**, context precision **0.74**.
- **Answer correctness (direct LLM judge): 91% correct, 98% correct-or-partial.**
- **Grounded / cited: 100% · Hallucinated: 0%** — every answer cites document · section · page, or correctly refuses.

3. Overall Performance on the Golden Set

Metric	Score
Retrieval accuracy (Top-1 / Top-5)	95% / 99%
Faithfulness (RAGAS)	0.96
Answer correctness (correct / correct-or-partial)	91% / 98%
Grounded / hallucinated	100% / 0%

4. Observations, Limitations, and Improvements

Strengths. Retrieval is strong and generation is highly faithful with 100% grounding — for a legal tool, *not hallucinating* is the critical property, and the system delivers it. Every answer is traceable to an exact document, section, and page. Graph RAG adds relationship queries ("which judgments cite the Fiscal Responsibility Act?") that semantic search cannot answer.

Limitations. Tax retrieval is the soft spot (84% Top-1) — IRS publications share heavy overlapping vocabulary, so the exact pub sometimes ranks 2nd–3rd (always within Top-5, so generation still succeeds). The corpus spans 2025 and 2026 tax years, so a year-agnostic question can surface both years' figures. Embeddings are general-purpose (bge-base-en-v1.5) rather than legal-domain-tuned.

Improvements. (1) Tax-year disambiguation in the generation prompt — label each figure with its year (implemented). (2) A legal-domain embedding model to lift Tax Top-1. (3) Broaden the citation-extraction recall in the graph to capture more cross-references.